

Gradient Derivation

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1 Neural Network overview

The neural network consists of three layers: the input layer, the hidden layer, and the output layer. The structure is

- **Input Layer:** The network takes in two input features x_1 and x_2 . For easy addition of bias, we include an extra constant input equal to 1 to the network making the total input dimension 3.
- **Hidden Layer:** The hidden layer has a total of 3 neurons, taking their inputs from the 3 input nodes. The weights for the inputs of the hidden layer are stored as a weight matrix $W^{(1)} \in \mathbb{R}^{3 \times 3}$. The output of each hidden neuron is computed as:

$$Z_j = \sigma(H_j), \quad j \in \{1, 2, 3\} \quad (1.1)$$

where σ is the sigmoid activation function.

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

and H is the input matrix for the hidden layer.

$$H_j = \sum_{m=1}^3 X_m W_{m,j}^{(1)}$$

- **Output Layer:** The output layer is a single neuron that receives inputs from all the 3 neurons of the hidden layer and an additional term for bias as earlier, (making it 4 inputs in total). The weights for the inputs of the output neuron are stored as another weight matrix $W^{(2)} \in \mathbb{R}^{4 \times 1}$. The input to the final layer is calculated as :

$$O = \sum_{k=1}^4 Z_k W_{k,1}^{(2)} \quad (1.2)$$

Here, $Z_4 = 1$ (for the bias term). The final output $\hat{y}$ of the neural network is obtained by applying the sigmoid activation function:

$$\hat{y} = \sigma(O) = \frac{1}{1 + e^{-O}} \quad (1.3)$$

This overview of the neural network gives us the basic understanding of the Neural Network.

2 Partial Derivative of $\mathcal{L}$ wrt $W_{k,1}^{(2)}$

We will compute the gradient of the loss function $\mathcal{L}$ with respect to the weight $W_{k,1}^{(2)}$ using the given expression.

$$\frac{\partial \mathcal{L}}{\partial W_{k,1}^{(2)}} = \sum_{i=1}^N \frac{\partial \mathcal{L}}{\partial \hat{y}_i} \times \frac{\partial \hat{y}_i}{\partial O_i} \times \frac{\partial O_i}{\partial W_{k,1}^{(2)}}, \quad k \in \{1, 2, 3, 4\} \quad (2.1)$$

2.1 Calculating $\frac{\partial \mathcal{L}}{\partial \hat{y}_i}$

The loss function is given by:

$$\mathcal{L} = \frac{1}{2N} \sum_{i=1}^N (Y_i - \hat{y}_i)^2$$

Taking its partial derivative wrt $\hat{y}_i$:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \hat{y}_i} &= \frac{\partial}{\partial \hat{y}_i} \left(\frac{1}{2N} \sum_{i=1}^N (Y_i - \hat{y}_i)^2 \right) \\ &= \frac{1}{2N} \times 2(Y_i - \hat{y}_i) \times (-1) \\ &= -\frac{1}{N} (Y_i - \hat{y}_i) \\ \frac{\partial \mathcal{L}}{\partial \hat{y}_i} &= \frac{1}{N} (\hat{y}_i - Y_i) \end{aligned} \quad (2.2)$$

2.2 Calculating $\frac{\partial \hat{y}_i}{\partial O_i}$

We have:

$$\hat{y}_i = \sigma(O_i) = \frac{1}{1 + e^{-O_i}}$$

its derivative is:

$$\frac{\partial \hat{y}_i}{\partial O_i} = \sigma'(O_i) \quad (2.3)$$

2.3 Calculating $\frac{\partial O_i}{\partial W_{k,1}^{(2)}}$

We have:

$$O_i = \sum_{j=1}^4 Z_{i,j} W_{j,1}^{(2)}$$

Taking the derivative:

$$\frac{\partial O_i}{\partial W_{k,1}^{(2)}} = Z_{i,k} \quad (2.4)$$

2.4 Final Expression (partial derivative w.r.t. $W_{k,1}^{(2)}$)

Using 2.2, 2.3, 2.4, in 2.1

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial W_{k,1}^{(2)}} &= \sum_{i=1}^N \left(\frac{1}{N} (\hat{y}_i - Y_i) \right) \times \sigma'(O_i) \times Z_{i,k} \\ &= \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - Y_i) \times \sigma'(O_i) \times Z_{i,k} \end{aligned}$$

This can be expressed in matrix operations as

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial W_{k,1}^{(2)}} &= \text{gradient2}[k] = \frac{1}{N} Z_k^\top ((\hat{y} - Y) \cdot \sigma'(O)) \\ \text{gradient2} &= \frac{1}{N} Z^\top ((\hat{y} - Y) \cdot \sigma'(O)) \end{aligned}$$

This can be calculated in python using the following code snippet

```
gradient2 = (Z.T @ ((predicted_y - training_y) * grad_sigmoid(0))) / N
```

3 Partial Derivative of $\mathcal{L}$ wrt $W_{k,l}^{(1)}$

We compute the partial derivative of the loss function L with respect to $W_{k,l}^{(1)}$ as follows:

$$\frac{\partial \mathcal{L}}{\partial W_{k,l}^{(1)}} = \sum_{i=1}^N \frac{\partial \mathcal{L}}{\partial \hat{y}_i} \cdot \frac{\partial \hat{y}_i}{\partial O_i} \cdot \frac{\partial O_i}{\partial Z_{i,l}} \cdot \frac{\partial Z_{i,l}}{\partial H_{i,l}} \cdot \frac{\partial H_{i,l}}{\partial W_{k,l}^{(1)}} \quad (3.1)$$

Solving part-by-part:

1.

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \hat{y}_i} &= \frac{\partial}{\partial \hat{y}_i} \left(\frac{1}{2N} \sum_{i=1}^N (Y_i - \hat{y}_i)^2 \right) \\ &= \frac{1}{N} (\hat{y}_i - Y_i) \end{aligned} \quad (3.2)$$

2.

$$\begin{aligned} \frac{\partial \hat{y}_i}{\partial O_i} &= \frac{\partial}{\partial O_i} (\sigma(O_i)) \\ &= \sigma'(O_i) \end{aligned} \quad (3.3)$$

3.

$$\begin{aligned} \frac{\partial O_i}{\partial Z_{i,l}} &= \frac{\partial}{\partial Z_{i,l}} \left(\sum_{j=1}^4 Z_{i,j} W_{j,1}^{(2)} \right) \\ &= W_{l,1}^{(2)} \end{aligned} \quad (3.4)$$

4.

$$\begin{aligned} \frac{\partial Z_{i,l}}{\partial H_{i,l}} &= \frac{\partial}{\partial H_{i,l}} (\sigma(H_{i,l})) \\ &= \sigma'(H_{i,l}) \end{aligned} \quad (3.5)$$

5.

$$\begin{aligned} \frac{\partial H_{i,l}}{\partial W_{k,l}^{(1)}} &= \frac{\partial}{\partial W_{k,l}^{(1)}} \left(\sum_{m=1}^3 X_{i,m} W_{m,l}^{(1)} \right) \\ &= X_{i,k} \end{aligned} \quad (3.6)$$

where $X_{i,k}$ is the k^{th} component of the input to the neural network for the i^{th} data point.

3.1 Final Expression (partial derivative w.r.t. $W_{k,l}^{(1)}$)

Using 3.2, 3.3, 3.4, 3.5, 3.6 in 3.1

$$\frac{\partial \mathcal{L}}{\partial W_{k,l}^{(1)}} = \sum_{i=1}^N \frac{1}{N} (\hat{y}_i - Y_i) \cdot \sigma'(O_i) \cdot W_{l,1}^{(2)} \cdot \sigma'(H_{i,l}) \cdot X_{i,k}$$

This can be expressed in matrix operations as:

$$\text{gradient1} = \frac{1}{N} X^\top \left[(\hat{y} - Y) \cdot \sigma'(O) \cdot W^{(2)} \cdot \sigma'(H) \right]$$

This can be calculated in python using the following code snippet

```
gradient1 = (training_x.T @ (((predicted_y - training_y) *  
↪ grad_sigmoid(O))[:,None] * grad_sigmoid(H) * weights2[:,-1])) / N
```